

DIFFPYRAMID: One Shot Generation for Text-to-Image Diffusion Models

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1. Introduction

Text-to-Image Diffusion Models have revolutionized generative AI by enabling image generations from textual descriptions. However, personalizing these models on a small data set remains a challenge. Existing methods, such as DreamBooth[6], rely on multiple reference images (often 3 to 5) for fine-tuning, which limits accessibility for users who only have a single reference image. This dependency creates a significant barrier to personalized image generation with minimal dataset, hindering its broader adoption.

To address this limitation, we propose DIFFPYRAMID, a one-shot learning approach that enables fine-tuning of Text-to-Image Diffusion Models. Our method tackles two key problems: the lack of sufficient data diversity from a single reference image and the need to generate high-quality, personalized outputs without sacrificing fidelity. Solving these problems broadens the application of generative AI.

DIFFPYRAMID combines text augmentation and image augmentation to overcome these challenges. By diversifying prompts through dependency parsing and conceptual embedding substitution, and enhancing image variability using stable diffusion and Laplacian pyramid blending, our method significantly expands the model’s understanding of the reference image and text.

Our contributions are threefold: first, we propose a novel pipeline for one-shot fine-tuning that eliminates the dependency on multiple reference images. Second, we introduce a data augmentation framework to enrich both textual and visual inputs, improving the quality and diversity of generated outputs. Also, we demonstrate the effectiveness of DIFFPYRAMID in achieving results comparable to multi-image fine-tuning. By lowering the barriers to customization, DIFFPYRAMID expands the applications of generative AI and democratize its use for a larger user group.

2. Related work

Text-to-Image Diffusion Models: Diffusion models have revolutionized text-to-image synthesis[5, 1], with DreamBooth leading the way in subject-driven generation by link-

ing unique identifiers to a model’s semantic space and using class-specific prior preservation loss to prevent overfitting. DreamBooth[6] excels in generating high-fidelity and contextually rich creative images using only 3–5 reference images, but still struggles when faced with extreme data scarcity, such as the one-shot generation setting where only a single reference image is available. Our method builds upon DreamBooth, adapting it to tackle the one-shot generation challenge by improving its ability to generate high-quality, personalized outputs with just one image as input.

Data Augmentations with Diffusion: Data augmentation is a widely used approach to mitigate data insufficiency. Beyond basic techniques like image manipulation[7], erasing[9], and mixup[8], advanced methods now leverage diffusion models for dataset augmentation. Recent works such as DIFFUSEMIX[3] and DIFFTPT[2] utilize diffusion-based generative processes to enhance data diversity. DIFFUSEMIX enriches single input images through generation, mask concatenation, and blending, producing highly diverse outputs at the cost of realism. In contrast, DIFFTPT employs a cosine similarity-based filtration to discard spurious augmentations, striking a balance between diversity and fidelity. Our approach combines the strengths of these methods by preserving the architecture and explores prompt augmentation, which further ensures the generation of high-quality outputs in text-to-image task settings.

3. Method

Our one-shot model introduces DIFFPYRAMID, a novel data augmentation technique designed to enhance the performance of DreamBooth[6] in one-shot settings. The two key components of DIFFPYRAMID are: generating a diverse and realistic set of augmented images from a single reference image and employing prompt augmentation to enrich textual inputs. Figure 1 illustrates the multi-level augmentation pipeline of DIFFPYRAMID.

For image augmentation, we start from a single reference image. We then apply stable diffusion with forwards and backwards to boost diversity. The outputs are blended with original image by Laplacian Pyramid, and a filter is used to

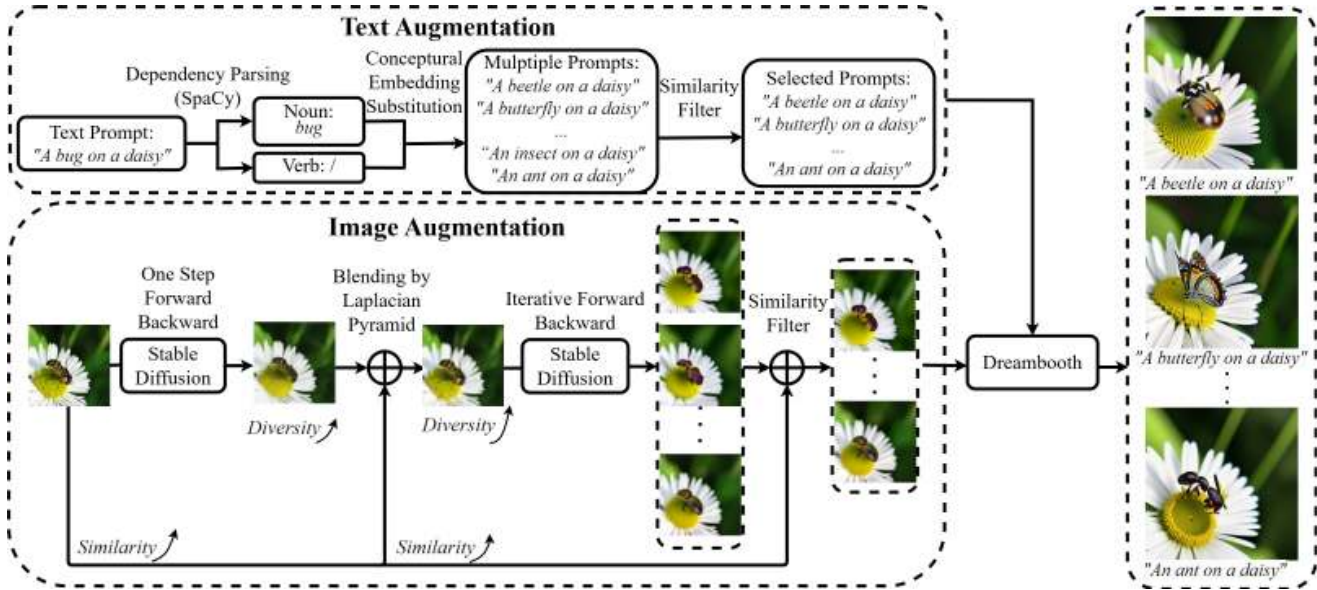


Figure 1. Architecture of the proposed DIFFPYRAMID approach.

ensure similarity between the augmented image set and the original image.

For text augmentation, we first apply dependency parsing methods to extract the subjects and actions from the text prompt. These key words are then replaced by their hyponyms. A similarity filter at the end of the pipeline ensures that the augmented prompts are more detailed than original ones, yet still preserving the same meaning.

Combining the above two pipelines, we get a bunch of different prompts and a set of input images. We can then run the Dreambooth model once on each prompt, and combine their outputs as the result of our image editing tasks.

4. Experiments

Dataset. The data used in this project is sourced from the publicly available *Flowers Recognition Dataset*[4]. To evaluate our method, we selected representative images from the dataset to test its effectiveness in generating high-quality, personalized outputs in a one-shot setting.

Hyperparameters. All input images in this project are resized to 512×512 . The Laplacian pyramid used in our method consists of 5 levels. For each original image, 8 augmented images are generated and fine-tuned with DreamBooth. Fine-tuning is performed with a learning rate of 10^{-6} over 2,000 training epochs. This setup ensures a consistent and fair evaluation across methods.

Baseline Methods. We compare our algorithm against two baseline methods: repeating and cropping. The repeating baseline involves duplicating the original image without alteration, while the cropping baseline employs random crops as a common data augmentation technique to extract por-

tions of the original image.

Evaluation Metrics. Two key criteria are used to evaluate our model: similarity to the original image and diversity among the generated images. For similarity, we utilize Frechet Inception Distance (FID), cosine similarity, and the Structural Similarity Index Measure (SSIM). For diversity, we compute the average cosine similarity between pairs of generated images to quantify variation.

Results. Table 1 summarizes the quantitative results, demonstrating that our algorithm consistently outperforms the baseline methods across all evaluation metrics. This indicates that our approach achieves superior similarity to the original images while simultaneously maintaining higher diversity in the generated outputs.

Method	FID	Cos Similarity	SSIM	Cos Diversity
DIFFPYRAMID	340.45	0.9209	0.6097	0.8657
Cropping	351.32	0.9121	0.5877	0.8875
Repeating	343.29	0.9158	0.5891	0.9036

Table 1. Comparison between DIFFPYRAMID and baseline methods performance.

5. Conclusions

In this project, we proposed the effective DIFFPYRAMID model, achieving superior performance across metrics compared to baselines by balancing contextual fidelity and diversity. While reproducible through shared datasets and code, performance may vary across different trials. Future work could enhance adaptability and generalization to multimodal datasets.

References

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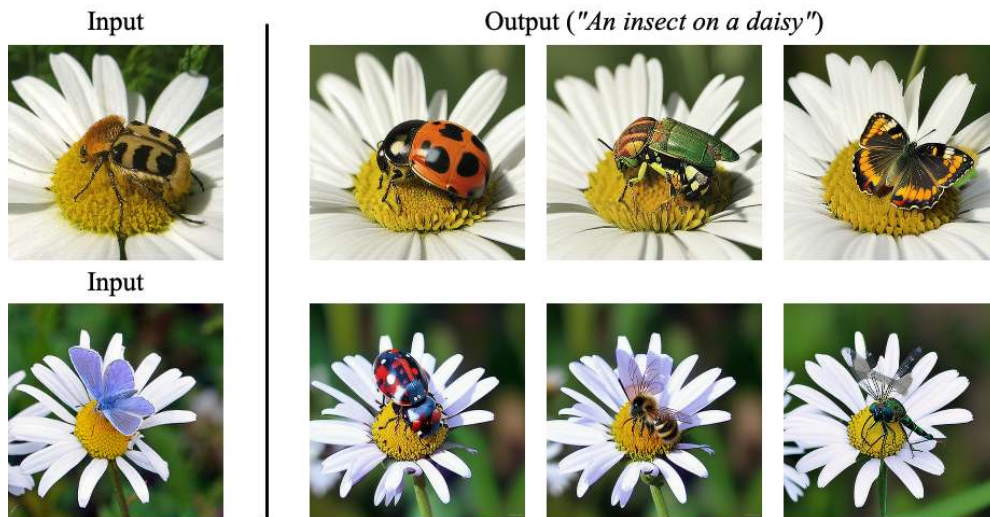


Figure 2. **Additional image-editing samples of the same class.** We are able to generate diverse species of insects while maintaining contextual consistency and high fidelity.

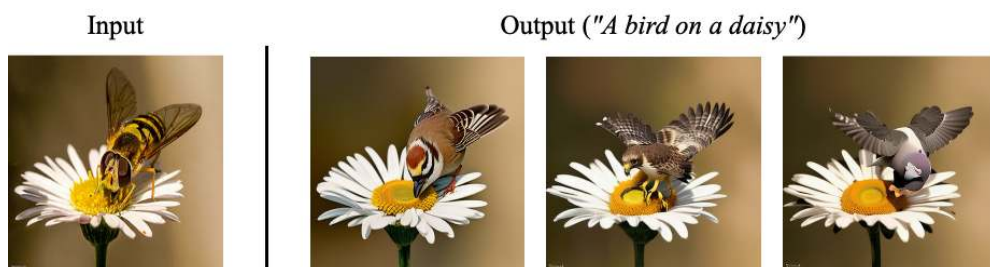


Figure 3. **Image-editing samples of different class.** We demonstrate our approach's ability to generate different birds from an image of insect while maintaining contextual consistency and sufficient fidelity.

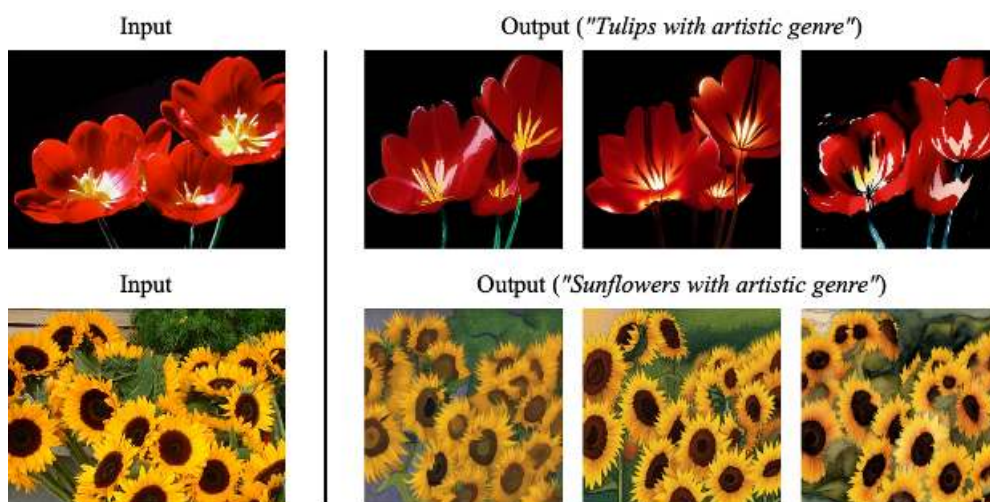


Figure 4. **Artistic-rendering samples of flowers.** The proposed approach transforms the artistic style of images from a realistic depiction to oil painting, cartoon, and ink wash painting styles, achieving both diversity and contextual consistency.